

Metaverse Track

Computer Programming

(Textbook INTRODUCTION TO PROGRAMMING USING PYTHON by Daniel Liang)

- Use of basic data type: Boolean, Int, Float, List, Tuple, String, Dictionary, Set
- Use of statement: selection, iteration
- Functions
- Classes and Objects
- Inheritance
- Polymorphism
- Exception Handling

Software Engineering Principles

(Textbook Software Engineering, 8th Edition by Ian Sommerville)

- Functional requirement
- Non-functional requirement
- User requirement
- System requirement

(Textbook PRACTICAL OBJECT-ORIENTED DESIGN WITH UML)

- 3-tier architecture
- Design of UML diagrams :
 - Use case diagram
 - Class diagram
 - Statechart diagram
- Implementation of class diagram and statechart in Python (Chapter 13)

Database Systems

- Database Systems Concepts
- Relational Database Model
- Entity Relationship Model

AI Track (A calculator is permitted in this exam)

Introduction to Logic

- Satisfiability, validity and logical equivalence in Propositional Logic
- Natural deduction in Propositional Logic
- Syntax and semantics of First-Order Logic (including translation between statements in English and First-Order Logic)

Linear Algebra

- Vectors
- Matrices
- Solution to systems of linear equations

Probability and Statistics

- Basic probability theory, Independent events, Bayes' Theorem
- Discrete random variables, Uniform and binomial probability distribution
- Continuous random variables, Uniform and normal probability distribution

Industrial IoT Track

Digital Circuits

Book: Digital Design (by M. Morris Mano and Michael D. Ciletti)

- Boolean algebra and logic gates
 - Boolean functions
 - Canonical and standard forms
 - Digital logic gates
- Combination logic
 - Combinational circuits
 - Binary adder-subtractor
 - Decoders / Encoders

Computer Architecture

Book: Structured Computer Organization, sixth edition (by Andrew S. Tanenbaum and Todd Austin, Pearson)

- Computer systems organization
 - Processors
 - Primary memory
 - Secondary memory
 - Input / Output
- Microarchitecture level
 - Design of the microarchitecture
- Parallel computer architectures
 - Instruction-level parallelism
 - Multithreading
 - Pipelining

Computer Networks

- Networking Fundamentals:

- OSI Model and TCP/IP: Understanding the layers of communication and core protocols.
- Basic Network Devices: Familiarity with routers, switches, hubs, and firewalls.
- Network Protocols: Knowledge of common protocols like Ethernet, IP, TCP, UDP.
- Basic Network Device Configuration: Ability to configure basic settings on network equipment.
- Routing and Switching:
 - Routing Protocol Configuration: Understanding protocols like RIP, OSPF, and LinkStage.
 - LAN Design and Planning: Ability to design and plan a local network.
 - Troubleshooting Routing and Switching Issues: Skills to diagnose and resolve routing and switching problems.
- Network Security:
 - Security Fundamentals: Understanding basic security principles like access control and encryption.
 - Firewalls and VPNs: Knowledge of firewalls and virtual private networks.
 - Malware and Intrusion Detection: Familiarity with malware protection and intrusion detection systems.

About the Track Selection Exams

- Exam schedule: **Sunday 30 June 2024, 09.30-17.00**

Time	Exam Paper
09.30-11.30	AI Track Exam , covering 3 subjects: ● Introduction to Logic (3 questions, 30-40 minutes) ● Probability and Statistics (3 questions, 30-40 minutes) ● Linear Algebra (3 questions, 30-40 minutes)
11.30-12.30	Lunch break
12.30-14.30	Metaverse Track Exam , covering 3 subjects: ● Computer Programming (3 questions, 30-40 minutes) ● Database Systems (3 questions, 30-40 minutes) ● Software Engineering Principles (3 questions, 30-40 minutes)
14.30-15.00	Break
15.00-17.00	Industrial IoT Track Exam , covering 3 subjects: ● Digital Circuits (3 questions, 30-40 minutes) ● Computer Architecture (3 questions, 30-40 minutes) ● Computer Networks (3 questions, 30-40 minutes)

- Each of the three exam papers consists of three parts, one part for each subject. Each part consists of 3 questions ~~with equal scoring weight~~ **(Correction: The 3**

parts/subjects in each exam paper are of equal weight, but the 3 exam questions in each part/subject may have different scoring weights). Each question may contain sub-questions.

- The exams are closed-book. Students are not permitted to bring any book or document. Calculators may be used in the AI Track Exam.
- The total time for each exam paper is 2 hours. So, the students are expected to spend 30-40 minutes on each of the three parts in the exam.
- Every student is required to take all the 3 exams.